INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

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(Affiliated to UEM Kolkata and Approved by AICTE)

Degree: BCA, Stream: BCA

Term – I , First Semester Examination, September - 2024

Subject Code: BCACC103

Subject Name: Discrete Structure

Full Marks: 30 Duration: 1 Hour

Part - A

Attempt 5 questions

Each question carries 2 marks (2 × 5)

1. If A={1,2,3,4,5,6,7,8,9}, B={2,3,4,5}, C={2,4,6,8} and D={4,5,6,7}, find B ∪ C and B ∩ D. **2**

or

If $U = \{x : x \in Z \text{ and } 1 \le x \le 10\}$ and $A = \{x : x \in U \text{ and } x \text{ is a prime number}\}$, $B = \{x : x \in U \text{ and } x \text{ is even}\}$, then find A Δ B. **2**

2. If U={-1,-2,0,3,5,10,12,13,16}, P={-2,3,5,12}, Q={-1,-2,0,5,12,13} then verify $(P \cap Q)' = P' \cup Q'$. **2**

or

If A={1,2,3,4,5,6,7,8,9}, B={2,3,4,5}, C={2,4,6,8} and D={4,5,6,7}, Verify that $(B \cup C) \cup (A \cup D) = A$ **2**

3. If A={1,2,3}, B={2,3,4}, write down the following set $\{(x, y) : (x, y) \in A \times B \text{ and } x < y\}$ **2**

or

If A={1,2,3}, B={2,3,4}, write down the following set $\{(x, y) : (x, y) \in A \times B \text{ and } x \ge y\}$ **2**

4. Given that A={1,2,3,5,7,9}, B={1,2,4,7,8} and C={1,2,3,5,7,9}, find (A ∪ B)-C. **2**

or

Find A ∪ (BΔC) when A={1,2,3,4,5}, B={1,2,4,8}, C={1,2,3,5,7}. **2**

5. If A={1,2}, B={2,3}, C={3,4}, then find $(A \times B) \cup (A \times C)$ **2**

or

If A={1,2}, B={2,3}, C={3,4}, then find $(A \times B) \cap (A \times C)$ **2**

Part - B

Attempt 2 questions

Each question carries 5 marks (5 × 2)

6. Let A,B,C be subsets of a Universal set U. Prove that **5**

$A \times (B \cup C) = (A \times B) \cup (A \times C)$

or

Let A,B,C be subsets of a Universal set U. Prove that **5**

$A \times (B - C) = (A \times B) - (A \times C)$

7. Show that the mapping f is injective but not surjective $f : N \to N$ defined by f(x)=x+1, $x \in N$ **5**

or

Show that the mapping f is injective but not surjective $f : Z \to Q$ defined by $f(x) = 2^x, x \in Z$ **5**

Part - C
Attempt 1 question
Each question carries 10 marks (10 × 1)

8. In a survey of 320 persons, number of persons taking tea is 210, taking milk 100 and coffee is 70. Number of persons who take tea and milk is 50, milk and coffee is 30, tea and coffee is 50. The number of persons taking all the three together is 20. Summarize the value which gives the number of persons who take neither tea nor milk nor coffee. **10**

or

In a survey concerning the smoking habits of consumers it was found that 55% smoke cigarette A, 50% smoke cigarette B, 42% smoke C, 28% smoke A and B, 20% smoke A and C, 12% smoke B and C and 10% smoke all the three cigarettes. Derive: **10**

i. What percentage do not smoke?

ii. What percentage smoke exactly 2 brands of cigarettes?